

UNIVERSITY OF EASTERN AFRICA, BARATON

SCHOOL OF BUSINESS

DEPARTMENT OF INFORMATION SYSTEM AND COMPUTING

PROJECT DONE IN PARTIAL FULFILMENT OF COURSE CODE;

COSC391

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DATA MINING PROJECT PROPOSAL

PROJECT TITLE: Detection of Fraudulent Mobile Money Transactions Using Data Mining Techniques

1. **Background of the problem and problem statement**
   i. Mobile money services have become part of our everyday lives, especially in Kenya, where many people depend on services such as M-Pesa to send and receive money, pay bills and carry out business transactions. These services have made sending and receiving money easier, faster and more convenient. However, as the number of people using mobile money continues to increase, cases of fraud have also become a concern. Some people lose money through unauthorized transactions, suspicious

money transfers, account misuse and other fraudulent activities. This affects not only the customers who lose their money but also the trust people have in mobile money services.

ii. The main problem addressed is how to identify fraudulent transactions before they cause financial losses. Every day, many mobile money transactions take place, making it difficult to manually check each one and determine whether it is genuine or suspicious. Although financial service providers use security measures to detect fraud, some fraudulent transactions may still go unnoticed, especially when fraudsters use new methods. The project will use data mining to study transaction records and identify patterns that may indicate fraudulent activities. By training a machine learning model using previous transaction data, we aim to develop a system that can predict whether a transaction is likely to be fraudulent or legitimate. This can help financial service providers identify suspicious activities mor quickly and take appropriate action.

2. Methodology

The main objective of this project is to develop a machine learning model that can detect potentially fraudulent mobile money transactions. We will use the **classification technique.** This has been chosen because our aim is to group transactions into two categories: **fraudulent** and **legitimate.** The model will learn from previous transaction records where the correct results are already known. After learning from this information, it will be able to examine new transactions and predict whether they may be fraudulent or genuine. The project will involve collecting the data, understanding its contents, cleaning and preparing it, training the model and testing how well it performs.

For our dataset, we will use the publicly available PaySim Synthetic Financial Dataset for Fraud Detection, which can be obtained from Kaggle at https://www.kaggle.com/datasets/ealaxi/paysim1. The dataset contains approximately 6,362,620 transaction records and includes details such as transaction type, amount transferred, account balances before and after transactions, and whether a transaction was fraudulent. The dataset sample to be used will be about 300,000 to 500,000 so as to keep computation manageable. The missing information, duplicate records, incorrect entries and other errors will be checked. Information such as transaction types will be converted into a format that the machine learning algorithms can understand. The data will then be divided into training, validation and testing sets so that we can train the model and check how well it works on transactions it has not seen before.

The machine learning algorithms to be used are:

i. **Logistic Regression-** Helps to establish a basic model for identifying fraudulent transactions

ii. **Decision Tree-** Helps understand the different conditions that may indicate suspicious activity.

iii. **Random Forest-** Combines several decision trees to improve the accuracy and reliability of predictions.

The three models will be trained and compared to their results to identify which one performs best. To measure their performance, accuracy, precision, recall, F-1 score and confusion matrix will be implemented. Particular attention will be given to precision and recall because detecting fraud is more important than simply achieving a high accuracy score. Validation process will help to select the best model before testing it using a separate

set of data. The model that gives the best balance between detecting fraudulent transactions and avoiding unnecessary fraud alerts will be selected.

3. **Main Objectives of the Project**
   The main objective of this project is to develop a data mining model that can help identify fraudulent mobile money transactions. The specific objectives are:
   a. To study mobile money transaction records and identify patterns that may be linked to fraud.
   b. To clean and prepare the collected data for use in machine learning.
   c. To develop and compare three classification algorithms: Logistic Regression, Decision Tree and Random Forest.
   d. To test the models and measure how accurately they can identify fraudulent transactions.
   e. To identify the best-performing model and explain how it can help improve the detection of mobile money fraud.
4. **Expected Outcomes**
   By the end of this project, we expect to have developed a machine learning model that can identify transactions that may be fraudulent based on the information provided in the dataset. We also expect to understand some of the common patterns associated with fraudulent mobile money activities and determine which of the three algorithms performs best. The project will demonstrate how data mining can be used to solve a real-life problem by helping financial service providers identify suspicious transactions and reduce the risk of financial losses. Although our dataset is simulated, the project will give us an opportunity to understand how similar technology could be applied in real mobile money services to improve transaction security.